

UTM FOUNDATION
2019/2020-3

FSPP0034 PHYSICS II

TEST 1

17 Jan 2020

ANSWER ALL THREE QUESTIONS

DURATION: 1 Hour

NAME:.....

IC NO:.....

LECTURER:.....

SECTION:.....

QUESTION 1 (10 MARKS)

- (a) i. State the Coulomb's law. (1 mark)

the electric force between two charges is proportional to the product of the charge and inversely proportional to the distance between them

- ii. Three electric charges $Q_1 = -76 \mu\text{C}$, $Q_2 = -55 \mu\text{C}$ and $Q_3 = 50 \mu\text{C}$ are placed each at the corners of a right triangle as shown in Figure 1 below. Determine the net electric force $\vec{F}_{\text{net}}$ acting on charge Q_1 .

$$F_{31} = \frac{9 \times 10^9 (50 \times 10^{-6})(76 \times 10^{-6})}{0.5^2} \quad (4 \text{ marks})$$

$$= 136.8 \text{ N}$$

$$F_{21} = \frac{9 \times 10^9 (55 \times 10^{-6})(76 \times 10^{-6})}{0.4^2} = 235.1 \text{ N}$$

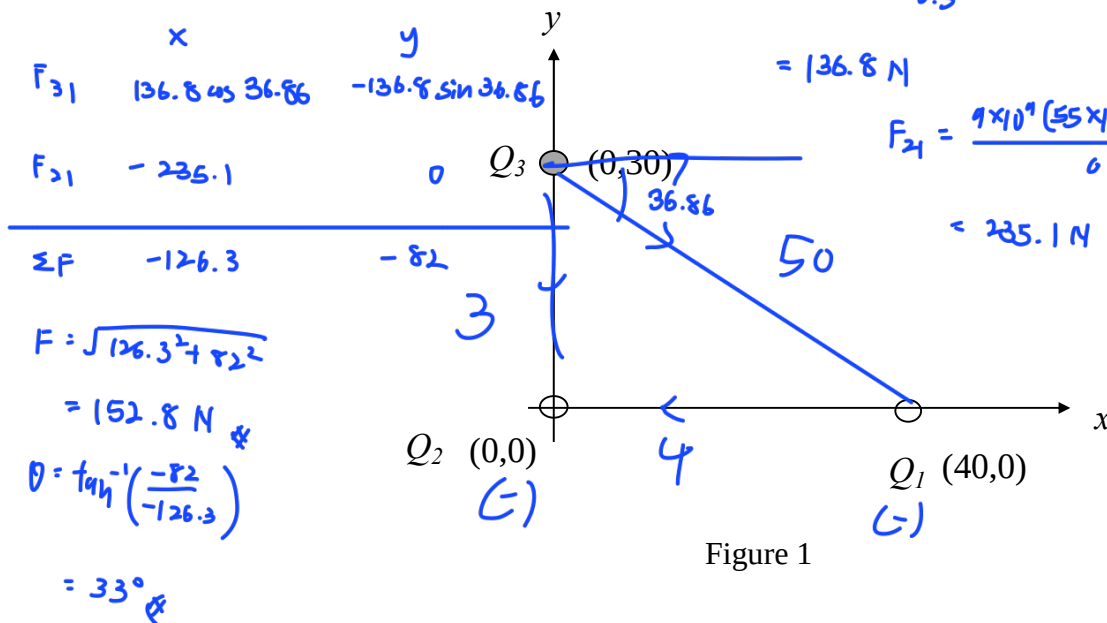


Figure 1

(b) Figure 2 shows two positive charges and two points (A and B). Calculate:

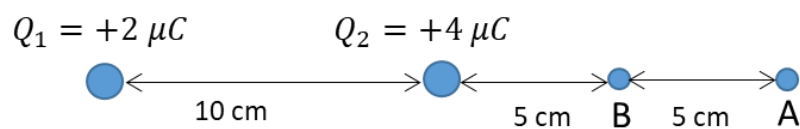


Figure 2

- i. Electric potential at point A and point B.

(3 marks)

$$V = \frac{U}{q^0}, \quad V = \frac{kQ}{r}$$

$$V_A = \frac{9 \times 10^9 (2 \times 10^{-6})}{0.20} + \frac{9 \times 10^9 (4 \times 10^{-6})}{0.10}$$

$$= 4.5 \times 10^5 \text{ V}$$

$$V_B = \frac{9 \times 10^9 (2 \times 10^{-6})}{0.15} + \frac{9 \times 10^9 (4 \times 10^{-6})}{0.05}$$

$$= 8.4 \times 10^5 \text{ V}$$

- ii. Potential difference between A and B.

$$\begin{aligned} V_{AB} &= V_A - V_B \\ &= 4.5 \times 10^{-6} - 8.4 \times 10^{-6} \\ &= -3.9 \times 10^{-6} \end{aligned}$$

(1 mark)

- iii. Work done to bring a charge $+1 \mu\text{C}$ from A to B.

$$V = \frac{W}{q_0}$$

$$W = -3.9 \times 10^{-6} (1 \times 10^{-6}) = -3.9 \times 10^{-12} \text{ J}$$

(1 mark)

QUESTION 2 (10 MARKS)

- (a) Suppose that the equivalent capacitance of three capacitors in Figure 3 is $2.77 \mu\text{F}$. Determine:

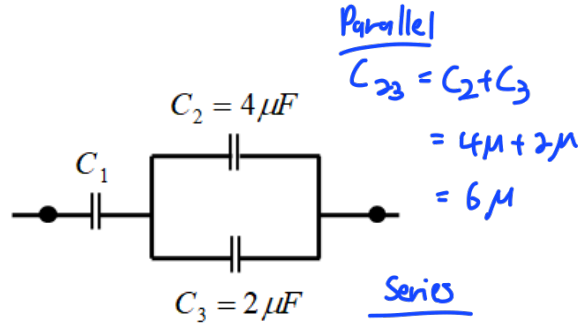


Figure 3

- i. The value for C_1 .

(3 marks)

ii. The voltage across C_1 if total charge is $12.6 \mu\text{C}$.

(1 mark)

$$Q_T = 12.6 \mu\text{C} = Q_1 = Q_2$$

$$Q_1 = C_1 V_1$$

$$12.6 \mu\text{C} = 5.146 \mu\text{F} (V_1)$$

$$V_1 = 2.45 \text{ V}$$

(b) A capacitor with plate area of $A = 600 \text{ cm}^2$ and plate separation $d = 20 \text{ mm}$ is connected to a 30 V battery. Calculate:

i. The capacitance of the capacitor.

(2 marks)

$$C = \frac{A \epsilon_0}{d} = \frac{600 \times 10^{-4} (8.85 \times 10^{-12})}{20 \times 10^{-3}} = 2.655 \times 10^{-11} \text{ F}$$

ii. The electric field.

(1 mark)

$$V = E d, \quad E = \frac{V}{d} = \frac{30}{20 \times 10^{-3}} = 1500 \text{ V m}^{-1}$$

iii. The energy stored in capacitor.

(1 mark)

$$U = \frac{1}{2} C V^2 = \frac{1}{2} (2.655 \times 10^{-11}) (1500)^2 \\ = 2.99 \times 10^{-5} \text{ J}$$

(c) If a piece of paper with dielectric constant $K = 2.7$ is now placed in between the two plates of capacitor in (b), determine:

i. The value of the capacitance.

(1 mark)

$$K = 2.7$$

$$2.7 = \frac{C}{2.655 \times 10^{-11}}$$

$$C = 7.1685 \times 10^{-11} \text{ C}$$

ii. The energy stored in the capacitor if the potential is 10 V.

(1 mark)

$$U = \frac{1}{2} CV^2$$

$$= \frac{1}{2} (7.1685 \times 10^{-11}) (10)^2$$

$$= 3.584 \times 10^{-9} \text{ J}$$

QUESTION 3 (10 MARKS)

(a) i. Define electric current I .

(1 mark)

electric current is the ^{nett} total charge flowing through area per unit time

$$I = \frac{Q}{t}$$

ii. A constant current of $8.0 \mu\text{A}$ flows through a piece of copper wire. How many electrons per minute pass through any point in that wire.

(2 marks)

$$Q = ne$$

$$\frac{Q}{t} = ne$$

$$\frac{8 \times 10^{-6}}{60} = n(1.6 \times 10^{-19})$$

$$n = 8.33 \times 10^{-27}$$

(b) i. Describe the difference between resistance R and resistivity ρ .

(1 mark)

ii. What is the length of a 0.50-mm diameter tungsten wire whose resistance is 0.57Ω ?

(Given: Resistivity of tungsten = $5.6 \times 10^{-8} \Omega\text{m}$)

(2 marks)

$$\mu = \frac{RA}{l}$$

$$l = \frac{RA}{\mu} = \frac{0.57(\pi \times 0.25 \times 10^{-4})}{5.6 \times 10^{-8}}$$

$$= 0.02\text{m} / 2\text{cm}$$

(c) i. State the Ohm's law for an ohmic device with resistance R .

- The Ohm Law state that the potential difference is proportional to the current flow through a conductor, temp is constant (1 mark)

- the graph of current I against V is a straight line through origin and has a gradient $\frac{1}{R}$.

ii. An ac voltage, whose peak value is 198 V, is applied across a 280- Ω resistor. What are the peak and rms currents, and average power dissipated in the resistor?

(3 marks)

$$V_0 = 198\text{V}$$

$$V_{\text{rms}} = \frac{198}{\sqrt{2}}$$

$$= 140\text{V}$$

$$V_{\text{rms}} = I_{\text{rms}} R$$

$$140 = I_{\text{rms}} (280)$$

$$I_{\text{rms}} = 0.5\text{A}$$

$$I_0 = 0.5 \times \sqrt{2}$$

$$= 0.707\text{A}$$

$$P = V_{\text{rms}} I_{\text{rms}}$$

$$= 140(0.5)$$

$$= 70\text{W}$$

APPENDIX

List of constant

$$e = -1.60 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$g = 9.80 \text{ m s}^{-2}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$\text{Coulomb's constant, } k = 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

List of formula

$$E = k \frac{Q}{r^2}$$

$$F = k \frac{Q_1 Q_2}{r^2}$$

$$U = k \frac{qQ}{r}$$

$$V = k \frac{Q}{r}$$

$$\Delta U = q \Delta V$$

$$\Delta V = -Ed$$

$$Q = CV$$

$$C_{eq} = C_1 + C_2 + C_3$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$E = K \frac{Q}{\varepsilon_o A}$$

$$U = \frac{1}{2} CV^2$$

$$P = \frac{U}{t}$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$\rho_T = \rho_o [1 + \alpha(T - T_o)]$$

$$R = \rho \frac{l}{A}$$

$$P = IV$$

$$I_{rms} = \frac{I_p}{\sqrt{2}}$$

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

$$\bar{P} = I_{rms} V_{rms}$$

$$I = \frac{Ne}{t}$$

$$j = \frac{I}{A}$$

$$v_d = \frac{j}{ne}$$